

# Lecture 5

## Lattices: Geometry, Hardness, and Reductions

SEC 595: Encrypted Computing

September 20, 2026

### Roadmap

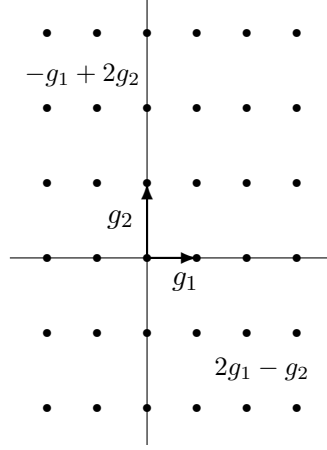
Lectures 1 through 4 built an algebraic machine:  $\mathbb{Z}_q$ , the ring  $R_q = \mathbb{Z}_q[X]/(X^n + 1)$ , the NTT, modules  $R_q^k$ , RNS and gadget decompositions, batching, automorphisms, embeddings. Every one of those tools is about *computing*. Not one says why anything should be *hard*.

This lecture supplies the missing half. The hardness is geometric: a discrete grid of points in high-dimensional space admits many bases, some far more useful than others, and converting a bad one into a good one appears to cost exponential time. Section 7 is the hinge — there an LWE sample becomes a point close to a lattice and an SIS solution a short vector in one. Sections 1–6 are preparation; Section 8 is a refinement. Each section states the property, says in a line where it is used, and works one small example completely; almost all of the examples live on a single 2-dimensional lattice.

One notational hazard. Lecture 1 fixed  $n$  as the *ring dimension*, the degree of  $X^n + 1$ ; most lattice texts use  $n$  for the lattice rank. Throughout Sections 1–7 we write  $d$  for the dimension of the ambient  $\mathbb{R}^d$  and  $k$  for the rank, so Minkowski’s bound reads  $\lambda_1 \leq \sqrt{d} \det(\mathcal{L})^{1/d}$  where other books write  $\sqrt{n}$ . In Section 8 both meanings are in play: a rank- $k$  module over a degree- $n$  ring is a lattice of dimension  $d = nk$ . And  $\beta$  is the SIS norm bound in Section 5, the BKZ block size from Section 6.3.

### 1. Lattices and Their Bases

A lattice is a regular, infinitely repeating grid of points. Choose a few arrows; the lattice is every point you can reach from the origin by walking a *whole number* of steps along each arrow, forwards or backwards.



**Definition 1** (Lattice, basis, rank, dimension). Let  $b_1, \dots, b_k \in \mathbb{R}^d$  be linearly independent over  $\mathbb{R}$ , and let  $B = (b_1 \mid \dots \mid b_k) \in \mathbb{R}^{d \times k}$ . The **lattice** generated by them is

$$\mathcal{L}(B) \stackrel{\text{def}}{=} \{ Bx : x \in \mathbb{Z}^k \} = \{ x_1 b_1 + \dots + x_k b_k : x_i \in \mathbb{Z} \}.$$

The vectors  $b_1, \dots, b_k$  form a **basis** of  $\mathcal{L}$  and  $B$  is its **basis matrix**;  $k$  is the **rank**,  $d$  the **dimension**, and  $\mathcal{L}$  is **full-rank** when  $k = d$ . Equivalently, a lattice is a discrete additive subgroup of  $\mathbb{R}^d$ .

**Example 1** (A basic lattice). Take  $d = k = 2$  with

$$g_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \quad g_2 = \begin{pmatrix} 0 \\ 3 \end{pmatrix}, \quad G = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix},$$

so that

$$\mathcal{L} = \mathcal{L}(G) = \{ x_1(2, 0) + x_2(0, 3) : x_1, x_2 \in \mathbb{Z} \} = 2\mathbb{Z} \times 3\mathbb{Z}.$$

This is the lattice drawn above. Some of its points:

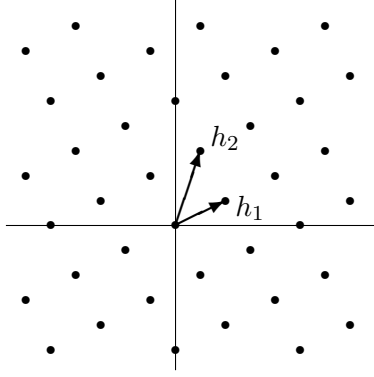
$$\begin{aligned} (2, 0) &= 1 \cdot g_1 + 0 \cdot g_2, \\ (4, -3) &= 2 \cdot g_1 - 1 \cdot g_2, \\ (-2, 6) &= -1 \cdot g_1 + 2 \cdot g_2, \\ (6, -9) &= 3 \cdot g_1 - 3 \cdot g_2. \end{aligned}$$

A point  $(u, v)$  lies in  $\mathcal{L}$  exactly when the coefficient vector  $x = G^{-1} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} u/2 \\ v/3 \end{pmatrix}$  is an integer vector:  $(6, -9)$  gives  $x = (3, -3) \in \mathbb{Z}^2$ , so it is in;  $(6, -8)$  gives  $x = (3, -8/3)$ , so it is not;  $(1, 0)$  gives  $x = (1/2, 0)$ , so it is not.

*A non-example.*  $b_1 = (1, 2)$  and  $b_2 = (2, 4) = 2b_1$  are linearly dependent, so they are not a basis of anything of rank 2: every combination  $x_1 b_1 + x_2 b_2 = (x_1 + 2x_2) b_1$  lies on one line, and the set generated is the rank-1 lattice  $\mathbb{Z} \cdot (1, 2)$ .

**Example 2** (Another lattice, not axis-aligned). Take

$$h_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad h_2 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \quad H = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \quad \mathcal{M} = \mathcal{L}(H) = \{ x_1(2, 1) + x_2(1, 3) : x_1, x_2 \in \mathbb{Z} \}.$$



Some of its points:

$$\begin{aligned} (3, 4) &= 1 \cdot h_1 + 1 \cdot h_2, \\ (4, 2) &= 2 \cdot h_1 + 0 \cdot h_2, \\ (0, 5) &= -1 \cdot h_1 + 2 \cdot h_2, \\ (-1, 2) &= -1 \cdot h_1 + 1 \cdot h_2. \end{aligned}$$

Here  $\det H = 2 \cdot 3 - 1 \cdot 1 = 5$ , so  $H^{-1} = \frac{1}{5} \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}$ , and  $(u, v)$  lies in  $\mathcal{M}$  exactly when  $x = H^{-1} \begin{pmatrix} u \\ v \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 3u-v \\ -u+2v \end{pmatrix}$  is an integer vector:  $(3, 4)$  gives  $x = \frac{1}{5}(5, 5) = (1, 1)$ , in;  $(1, 1)$  gives  $x = \frac{1}{5}(2, 1)$ , out. Equivalently,  $(u, v) \in \mathcal{M}$  if and only if  $u + 3v \equiv 0 \pmod{5}$  — check  $(3, 4)$ :  $3 + 12 = 15$ ;  $(1, 1)$ :  $1 + 3 = 4$ .

### 1.3 Two Bases, One Lattice

**Theorem 1.** Let  $B, \tilde{B} \in \mathbb{R}^{d \times k}$  have linearly independent columns. Then  $\mathcal{L}(B) = \mathcal{L}(\tilde{B})$  if and only if  $\tilde{B} = BU$  for some **unimodular**  $U \in \mathbb{Z}^{k \times k}$ , i.e. an integer matrix with  $\det U = \pm 1$ .

**Example 3** (Two bases of the same lattice). Start from  $\mathcal{L} = 2\mathbb{Z} \times 3\mathbb{Z}$  with basis  $G$  from Example 1, and take

$$U = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}, \quad \det U = 1 \cdot 3 - 2 \cdot 1 = 1, \quad U^{-1} = \begin{pmatrix} 3 & -2 \\ -1 & 1 \end{pmatrix} \in \mathbb{Z}^{2 \times 2}.$$

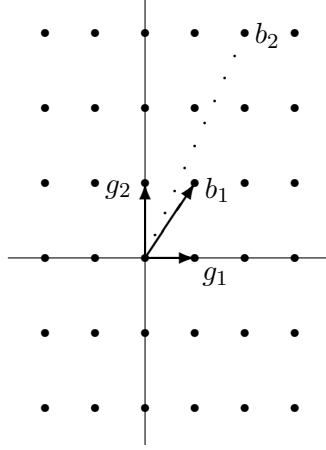
Then

$$B = GU = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 4 \\ 3 & 9 \end{pmatrix}, \quad b_1 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \quad b_2 = \begin{pmatrix} 4 \\ 9 \end{pmatrix}.$$

$B$  generates only lattice points:  $b_1 = g_1 + g_2$  and  $b_2 = 2g_1 + 3g_2$ , so every integer combination of  $b_1, b_2$  is an integer combination of  $g_1, g_2$ .  $B$  generates all of them: from  $G = BU^{-1}$ ,

$$g_1 = 3b_1 - b_2 = (6, 9) - (4, 9) = (2, 0), \quad g_2 = -2b_1 + b_2 = (-4, -6) + (4, 9) = (0, 3).$$

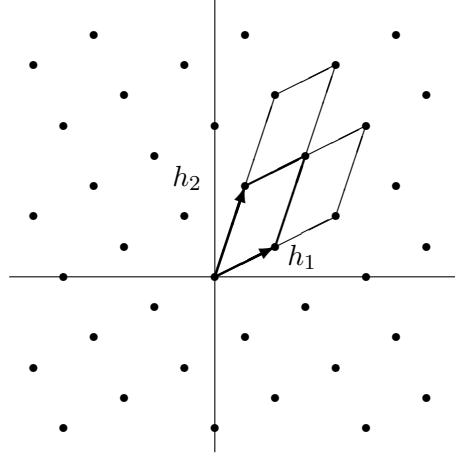
So  $\mathcal{L}(B) = \mathcal{L}(G)$ . ✓ The two bases drawn on the same points:



Solid arrows are  $g_1, g_2$ ; the oblique arrow and the dotted segment are  $b_1 = (2, 3)$  and  $b_2 = (4, 9)$ . Both pairs reach exactly the same dots. Finally  $\det G = 6$  and  $\det B = 2 \cdot 9 - 4 \cdot 3 = 6$ : the same number, which is Section 2.

## 2. The Determinant

A lattice tiles space with copies of a single cell, the parallelogram spanned by the basis vectors. The determinant is the area of that cell — the amount of room each lattice point has to itself.



The thick parallelogram is  $\mathcal{P}(H)$  for the lattice  $\mathcal{M}$  of Example 2, with vertices  $0, h_1, h_2$  and  $h_1 + h_2$ ; the thin ones are two of its translates by lattice vectors, and copies like them fill the plane with no gaps and no overlaps.

**Definition 2** (Fundamental parallelepiped and determinant). For linearly independent  $b_1, \dots, b_k$ , the **fundamental parallelepiped** is

$$\mathcal{P}(B) \stackrel{\text{def}}{=} \left\{ \sum_i x_i b_i : x_i \in \mathbb{R}, 0 \leq x_i < 1 \right\}.$$

For a full-rank lattice  $\mathcal{L} = \mathcal{L}(B)$  the **determinant**, or **covolume**, is

$$\det(\mathcal{L}) \stackrel{\text{def}}{=} |\det B| = \text{vol}(\mathcal{P}(B));$$

for rank  $k < d$ ,  $\det(\mathcal{L}) = \sqrt{\det(B^T B)}$ .

**Proposition 1.**  $\det(\mathcal{L})$  does not depend on the basis: two bases satisfy  $\tilde{B} = BU$  with  $|\det U| = 1$  by Theorem 1, hence  $|\det \tilde{B}| = |\det B|$ . If  $\mathcal{L} \subseteq \mathbb{Z}^d$  is full-rank then  $\det(\mathcal{L}) = [\mathbb{Z}^d : \mathcal{L}]$ , the number of cosets of  $\mathcal{L}$  in  $\mathbb{Z}^d$ ; a lattice cut out of  $\mathbb{Z}^d$  by  $k$  independent linear congruences modulo  $q$  therefore has  $\det(\mathcal{L}) = q^k$ .

**Example 4** (One lattice, two bases, one determinant — and the index). For  $\mathcal{L} = 2\mathbb{Z} \times 3\mathbb{Z}$  of Example 3,

$$\det(\mathcal{L}) = |\det G| = |2 \cdot 3 - 0| = 6, \quad |\det B| = |2 \cdot 9 - 3 \cdot 4| = 6. \checkmark$$

The parallelepiped  $\mathcal{P}(G)$  is the  $2 \times 3$  rectangle;  $\mathcal{P}(B)$  is the slanted parallelogram with vertices  $0, (2, 3), (4, 9), (6, 12)$ . Both have area 6.

Now the index. Membership in  $\mathcal{L}$  depends only on  $(u \bmod 2, v \bmod 3)$ , so the cosets of  $\mathcal{L}$  in  $\mathbb{Z}^2$  are represented by

$$(0, 0), \quad (1, 0), \quad (0, 1), \quad (1, 1), \quad (0, 2), \quad (1, 2),$$

which are six distinct classes, and every  $(u, v) \in \mathbb{Z}^2$  is congruent modulo  $\mathcal{L}$  to exactly one of them. Hence  $[\mathbb{Z}^2 : \mathcal{L}] = 6 = \det(\mathcal{L})$ .  $\checkmark$

**Example 5** (The slanted lattice: area of the cell, and the index). For  $\mathcal{M} = \mathcal{L}(H)$  of Example 2,

$$\det(\mathcal{M}) = |\det H| = |2 \cdot 3 - 1 \cdot 1| = 5,$$

so the parallelogram drawn in Section 2.1, with vertices  $0, (2, 1), (1, 3), (3, 4)$ , has area 5: the triangle  $0, h_1, h_2$  has area  $\frac{1}{2}|2 \cdot 3 - 1 \cdot 1| = \frac{5}{2}$ , and the parallelogram is two copies of it.  $\checkmark$

For the index, Example 2 showed that  $(u, v) \in \mathcal{M}$  exactly when  $u + 3v \equiv 0 \pmod{5}$ , so two integer points are congruent modulo  $\mathcal{M}$  exactly when  $u + 3v$  agrees modulo 5. The five values of  $u + 3v \bmod 5$  give five cosets, represented by

$$(0, 0), \quad (1, 0), \quad (2, 0), \quad (3, 0), \quad (4, 0),$$

and  $[\mathbb{Z}^2 : \mathcal{M}] = 5 = \det(\mathcal{M})$ .  $\checkmark$

Section 7 uses the last clause of Proposition 1: the lattice  $\Lambda_q^\perp(A)$  is cut out of  $\mathbb{Z}^d$  by  $k$  congruences modulo  $q$ , so  $\det \Lambda_q^\perp(A) = q^k$ .

### 3. Gram–Schmidt and Basis Quality

A basis is *good* when its vectors sit nearly at right angles to one another and *bad* when they lie nearly along one line. Gram–Schmidt strips from each vector the part it shares with the earlier ones; how little survives measures how far from orthogonal the basis is. The orthogonality defect compresses that into one number: 1 for a right-angled basis, large for a skewed one.

**Definition 3** (Gram–Schmidt orthogonalisation and orthogonality defect). For linearly independent  $b_1, \dots, b_k$  the **Gram–Schmidt orthogonalisation** (GSO) is

$$\tilde{b}_1 = b_1, \quad \tilde{b}_i = b_i - \sum_{j < i} \mu_{i,j} \tilde{b}_j, \quad \mu_{i,j} = \frac{\langle b_i, \tilde{b}_j \rangle}{\langle \tilde{b}_j, \tilde{b}_j \rangle},$$

so that  $B = \tilde{B}M$  with  $M$  upper triangular, 1's on the diagonal and the  $\mu_{i,j}$  off it. For a full-rank basis  $B$  the **orthogonality defect** is

$$\delta(B) = \frac{\prod_{i=1}^d \|b_i\|_2}{\det(\mathcal{L})}.$$

**Theorem 2.** For a full-rank lattice with basis  $b_1, \dots, b_d$ ,

$$\det(\mathcal{L}) = \prod_{i=1}^d \|\tilde{b}_i\|_2,$$

and consequently  $\delta(B) \geq 1$ , with equality exactly when  $b_1, \dots, b_d$  are pairwise orthogonal.

*Proof.*  $B = \tilde{B}M$  with  $\det M = 1$ , so  $|\det B| = |\det \tilde{B}|$ ; and  $\tilde{B}$  has pairwise orthogonal columns, so  $|\det \tilde{B}| = \prod_i \|\tilde{b}_i\|_2$ . Each  $\tilde{b}_i$  is  $b_i$  minus a projection, so  $\|\tilde{b}_i\|_2 \leq \|b_i\|_2$  termwise; multiplying these gives  $\det(\mathcal{L}) \leq \prod_i \|b_i\|_2$ , which is  $\delta(B) \geq 1$ , with equality exactly when no projection is subtracted.  $\square$

**Example 6** (A good and a bad basis of  $2\mathbb{Z} \times 3\mathbb{Z}$ , with both defects). Both bases of Example 3 generate  $\mathcal{L}$ , so both are measured against the same  $\det(\mathcal{L}) = 6$ .

The good basis  $g_1 = (2, 0)$ ,  $g_2 = (0, 3)$  is already orthogonal, so  $\tilde{g}_1 = g_1$ ,  $\tilde{g}_2 = g_2$ ,  $\mu_{2,1} = 0$ , and

$$\|\tilde{g}_1\|_2 \|\tilde{g}_2\|_2 = 2 \cdot 3 = 6 = \det(\mathcal{L}) \checkmark, \quad \delta(G) = \frac{2 \cdot 3}{6} = 1.$$

The bad basis  $b_1 = (2, 3)$ ,  $b_2 = (4, 9)$  has  $\|b_1\|_2 = \sqrt{13}$  and  $\|b_2\|_2 = \sqrt{97}$ , so

$$\delta(B) = \frac{\sqrt{13} \sqrt{97}}{6} = \frac{\sqrt{1261}}{6} \approx 5.918.$$

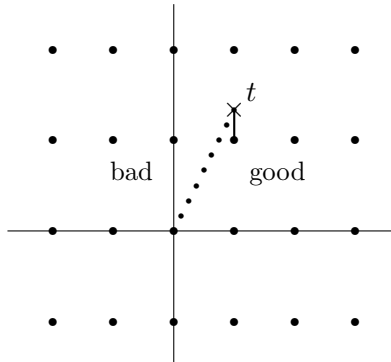
Its GSO is

$$\tilde{b}_1 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \quad \mu_{2,1} = \frac{\langle b_2, \tilde{b}_1 \rangle}{\|\tilde{b}_1\|_2^2} = \frac{8 + 27}{13} = \frac{35}{13}, \quad \tilde{b}_2 = \begin{pmatrix} 4 \\ 9 \end{pmatrix} - \frac{35}{13} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} -18/13 \\ 12/13 \end{pmatrix},$$

with  $\|\tilde{b}_1\|_2 = \sqrt{13} \approx 3.606$  and  $\|\tilde{b}_2\|_2 = \sqrt{468}/13 = 6/\sqrt{13} \approx 1.664$ , whose product is  $6 = \det(\mathcal{L})$ , as Theorem 2 requires  $\checkmark$ . The second vector shrank from  $\sqrt{97} \approx 9.849$  to 1.664 under orthogonalisation.

### 3.3 Babai Rounding

To decode a point of space to a nearby lattice point, write the point in the basis and round each coefficient to a whole number. With a right-angled basis the rounding lands on the truly nearest lattice point; with a skewed one it can land far away, because rounding follows the basis directions rather than the straight ones. A **lattice trapdoor** is exactly that asymmetry: the public key is a bad basis of  $\mathcal{L}$ , the secret key a good basis of the same  $\mathcal{L}$ , and decoding succeeds only with the good one.



The cross is the target  $t$ . The short solid link runs to the lattice point the good basis returns; the dotted link runs to the one the bad basis returns.

**Definition 4** (Babai rounding and the decoding radius). Let  $B$  be a basis of a full-rank lattice  $\mathcal{L}$  and  $t \in \mathbb{R}^d$  a target. **Babai rounding** returns  $B \lfloor B^{-1}t \rfloor$ : write  $t$  in the basis and round each coefficient to the nearest integer. Its error lies in  $\mathcal{P}(B)$  recentred at the origin. Babai's **nearest-plane** refinement returns the genuinely closest lattice vector to  $t$  whenever

$$\text{dist}(t, \mathcal{L}) < \frac{1}{2} \min_i \|\tilde{b}_i\|_2,$$

the right-hand side being the **decoding radius** of  $B$ .

**Example 7** (The same target decoded by both bases). Take  $\mathcal{L} = 2\mathbb{Z} \times 3\mathbb{Z}$  and the target  $t = (2, 4)$ , whose nearest lattice point is  $(2, 3)$ , at distance 1.

With the good basis  $G$ :  $G^{-1}t = (2/2, 4/3) = (1, 4/3)$ , and  $\lfloor 4/3 \rfloor = 1$ , so the output is

$$G \begin{pmatrix} 1 \\ 1 \end{pmatrix} = (2, 3), \quad \text{error } \|t - (2, 3)\|_2 = 1. \quad \checkmark$$

With the bad basis  $B$ :  $B^{-1} = \frac{1}{6} \begin{pmatrix} 9 & -4 \\ -3 & 2 \end{pmatrix}$ , so

$$B^{-1}t = \frac{1}{6} \begin{pmatrix} 9 \cdot 2 - 4 \cdot 4 \\ -3 \cdot 2 + 2 \cdot 4 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 1/3 \\ 1/3 \end{pmatrix}, \quad \lfloor 1/3 \rfloor = 0,$$

so the output is  $B \begin{pmatrix} 0 \\ 0 \end{pmatrix} = (0, 0)$ , at distance  $\|(2, 4)\|_2 = \sqrt{20} \approx 4.472$  from  $t$ . The bad basis misses by more than four times the true distance.

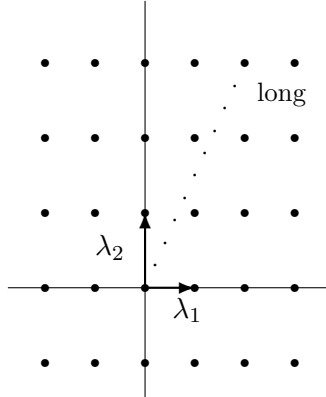
The decoding radii explain it. From Example 6,

$$\frac{1}{2} \min_i \|\tilde{g}_i\|_2 = \frac{1}{2} \cdot 2 = 1, \quad \frac{1}{2} \min_i \|\tilde{b}_i\|_2 = \frac{1}{2} \cdot \frac{6}{\sqrt{13}} \approx 0.832.$$

The target sits at distance 1 from  $\mathcal{L}$ : inside the good basis's radius, outside the bad one's.

## 4. How Short Is Short?

$\lambda_1$  is the length of the shortest nonzero vector in the lattice;  $\lambda_2$  is the shortest length at which two independent lattice vectors are available, and so on up. Minkowski's theorem caps  $\lambda_1$  in terms of the dimension and the determinant alone, and the Gaussian heuristic predicts what it typically is.



The two solid arrows are shortest vectors of their kind, of lengths 2 and 3; the dotted one runs to (4, 9), a lattice vector more than four times as long.

**Definition 5** (Successive minima). For a lattice  $\mathcal{L}$  of rank  $k$  and  $1 \leq i \leq k$ , the  $i$ -th **successive minimum**  $\lambda_i(\mathcal{L})$  is the least  $r > 0$  such that  $\mathcal{L}$  contains  $i$  linearly independent vectors of Euclidean norm at most  $r$ . In particular  $\lambda_1(\mathcal{L})$  is the length of a shortest nonzero lattice vector, and  $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_k$ . Vectors attaining the minima need not form a basis.

**Theorem 3** (Minkowski). *Let  $\mathcal{L} \subset \mathbb{R}^d$  be full-rank and  $S \subseteq \mathbb{R}^d$  convex, symmetric about the origin, with  $\text{vol}(S) > 2^d \det(\mathcal{L})$ . Then  $S$  contains a nonzero lattice point. Applying this to cubes gives **Minkowski's first theorem**,*

$$\lambda_1(\mathcal{L}) \leq \sqrt{d} \det(\mathcal{L})^{1/d}.$$

**Definition 6** (Gaussian heuristic). Assume the lattice points are spread like a Poisson process of density  $1/\det(\mathcal{L})$ , so that a region  $S$  holds about  $\text{vol}(S)/\det(\mathcal{L})$  of them. Setting that count to 1 for a ball of radius  $r$  gives the estimate

$$\lambda_1(\mathcal{L}) \approx \text{gh}(\mathcal{L}) = \frac{\det(\mathcal{L})^{1/d}}{V_d^{1/d}} \approx \sqrt{\frac{d}{2\pi e}} \det(\mathcal{L})^{1/d}, \quad V_d = \frac{\pi^{d/2}}{\Gamma(\frac{d}{2} + 1)},$$

the second form using  $V_d^{1/d} \approx \sqrt{2\pi e/d}$ , with constant  $\sqrt{1/2\pi e} \approx 0.2420$ . It is asymptotic, and unreliable in small dimension.

### 4.3 Example

**Example 8** (Minima, Minkowski and the heuristic, all as numbers). Take the running lattice  $\mathcal{L} = 2\mathbb{Z} \times 3\mathbb{Z}$ , with  $d = 2$  and  $\det(\mathcal{L}) = 6$ , so  $\det(\mathcal{L})^{1/d} = \sqrt{6} \approx 2.449$ .

*True minima.* A nonzero  $(u, v) \in \mathcal{L}$  with  $v = 0$  has  $|u| \geq 2$ ; any with  $v \neq 0$  has  $|v| \geq 3$ . Hence

$$\lambda_1 = 2 \text{ at } \pm(2, 0), \quad \lambda_2 = 3 \text{ at } \pm(0, 3),$$

and the good basis of Example 6 attains both.

*Minkowski.*  $\sqrt{2} \cdot \sqrt{6} = \sqrt{12} \approx 3.464$ , and  $2 \leq 3.464 \checkmark$ .

*Gaussian heuristic.* With  $V_2 = \pi$ ,

$$\text{gh}(\mathcal{L}) = \frac{\sqrt{6}}{\sqrt{\pi}} \approx 1.382, \quad \text{asymptotic form } \sqrt{\frac{2}{2\pi e}} \cdot \sqrt{6} \approx 0.838.$$

Both fall *below* the true  $\lambda_1 = 2$ , by factors of 1.45 and 2.39.

Section 6.4 uses the Gaussian heuristic, at dimensions in the hundreds where it is the standard estimate.

## 5. The Computational Problems

Throughout, the input is a basis  $B$  of  $\mathcal{L}$ : an arbitrary one, which in practice means a bad one.

- **SVP** asks for the shortest nonzero vector of the lattice.

- **CVP** asks, given a point of space, for the lattice vector nearest to it.
- **BDD** is CVP with a promise that the point is already close to the lattice, so the answer is unique.
- **SIS** asks, given a matrix modulo  $q$ , for a short nonzero integer vector that the matrix sends to zero.

**Definition 7** (SVP, CVP, BDD). Given a basis  $B$  of  $\mathcal{L}$ ,

- **SVP** (shortest vector): find nonzero  $v \in \mathcal{L}$  with  $\|v\|_2 = \lambda_1(\mathcal{L})$ .
- **CVP** (closest vector): given also a target  $t \in \mathbb{R}^d$ , find  $v \in \mathcal{L}$  minimising  $\|t - v\|_2$ .
- **BDD** (bounded distance decoding), parameter  $\alpha < 1/2$ : CVP with the *promise* that  $t$  satisfies  $\text{dist}(t, \mathcal{L}) < \alpha \lambda_1(\mathcal{L})$ , which makes the answer unique.

The relaxed forms are  $\gamma$ -SVP, asking for  $\|v\|_2 \leq \gamma \lambda_1$ ;  $\gamma$ -CVP, asking for  $\|t - v\|_2 \leq \gamma \text{dist}(t, \mathcal{L})$ ; SIVP $_\gamma$ , asking for  $d$  independent vectors of norm at most  $\gamma \lambda_d$ ; and the decision form GapSVP $_\gamma$ , which takes  $(B, r)$  and answers YES if  $\lambda_1 \leq r$ , NO if  $\lambda_1 > \gamma r$ , the gap excluded by promise.

What is known is a ladder in  $\gamma$ : exact SVP and CVP are NP-hard, and SVP stays NP-hard for constant  $\gamma$ ;  $\gamma = 2^{O(d)}$  is polynomial-time by LLL (Section 6); and at  $\gamma = \text{poly}(d)$ , where cryptography sits, there is neither an algorithm nor an NP-hardness proof, GapSVP $_\gamma$  lying in  $\text{NP} \cap \text{coNP}$  once  $\gamma \geq \sqrt{d/\log d}$ .

**Definition 8** (SIS). **Short Integer Solution**, parameters  $q, k, d, \beta$ . Given uniformly random  $A \in \mathbb{Z}_q^{k \times d}$  with  $d > k$ , find nonzero  $z \in \mathbb{Z}^d$  with

$$Az \equiv 0 \pmod{q} \quad \text{and} \quad \|z\|_2 \leq \beta.$$

The inhomogeneous version asks instead for short  $z$  with  $Az \equiv u$  for a given target  $u \in \mathbb{Z}_q^k$ . Both adjectives are needed: without “short”  $z = (q, 0, \dots, 0)$  works, without “nonzero”  $z = 0$  works, and  $\beta$  must stay below  $q$ . A nonzero  $z$  with  $\|z\|_\infty \leq 1$  exists by pigeonhole as soon as  $2^d > q^k$ , i.e.  $d > k \log_2 q$ ; the same counting makes  $z \mapsto Az \bmod q$  a compressing map whose collisions are SIS solutions.

### 5.3 Example

**Example 9** (A hand-sized SIS instance). Take  $q = 13$ ,  $k = 1$ ,  $d = 3$  and  $A = \begin{pmatrix} 1 & 5 & 8 \end{pmatrix} \in \mathbb{Z}_{13}^{1 \times 3}$ , so we want nonzero integer  $z = (z_1, z_2, z_3)$  with

$$z_1 + 5z_2 + 8z_3 \equiv 0 \pmod{13}.$$

No single coordinate works, since 1, 5 and 8 are all nonzero mod 13. But  $5 + 8 = 13$ , so

$$z = (0, 1, 1), \quad \|z\|_2 = \sqrt{2} \approx 1.414,$$

and exhaustive search over  $\|z\|_\infty \leq 6$  confirms that  $\pm(0, 1, 1)$  are the only solutions of norm  $\sqrt{2}$  and that none is shorter. So the instance is solvable for any  $\beta \geq \sqrt{2}$  and unsolvable for  $\beta < \sqrt{2}$ ; with  $\beta \geq 13$  it is trivial, since  $(13, 0, 0)$  is then a solution.

## 6. Lattice Reduction

Lattice reduction takes a bad basis and returns a good basis of the same lattice: it repeatedly subtracts multiples of one basis vector from another and swaps them when that makes an earlier

vector shorter, until no such move helps. The shorter the first output vector, the stronger the algorithm.

**Definition 9** (LLL-reduced basis, root Hermite factor, BKZ). A basis is **LLL-reduced** with parameter  $\delta \in (1/4, 1)$  when it is *size reduced*,  $|\mu_{i,j}| \leq \frac{1}{2}$  for all  $j < i$ , and satisfies the *Lovász condition*  $\delta \|\tilde{b}_{i-1}\|_2^2 \leq \|\tilde{b}_i + \mu_{i,i-1}\tilde{b}_{i-1}\|_2^2$  for  $i = 2, \dots, d$ ; when the second fails the algorithm swaps  $b_{i-1}$  and  $b_i$  and repeats. In dimension 2 this is **Lagrange–Gauss**: order so  $\|b_1\|_2 \leq \|b_2\|_2$ , replace  $b_2 \leftarrow b_2 - \lfloor \mu \rfloor b_1$  with  $\mu = \langle b_1, b_2 \rangle / \langle b_1, b_1 \rangle$ , then swap and repeat until  $b_2$  stops shrinking.

An algorithm whose output satisfies  $\|b_1\|_2 = \delta_0^d \cdot \det(\mathcal{L})^{1/d}$  achieves **root Hermite factor**  $\delta_0$ ; smaller is stronger.

**BKZ** with block size  $\beta$  runs LLL, then repeatedly solves exact SVP in the  $\beta$ -dimensional projected sublattices spanned by  $\tilde{b}_i, \dots, \tilde{b}_{i+\beta-1}$ , inserts the short vector found, and re-reduces until nothing improves;  $\beta = 2$  is LLL and  $\beta = d$  solves SVP exactly. One SVP call in dimension  $\beta$  costs  $2^{0.292\beta}$  classically and  $2^{0.265\beta}$  quantumly, and the quality reached is estimated by

$$\delta_0(\beta) \approx \left( \frac{\beta}{2\pi e} (\pi\beta)^{1/\beta} \right)^{1/(2(\beta-1))}.$$

**Theorem 4** (Lenstra–Lenstra–Lovász, 1982). *With  $\delta = 3/4$  the algorithm terminates in time polynomial in  $d$  and in the input bit-length, and returns a basis with*

$$\|b_1\|_2 \leq 2^{(d-1)/2} \lambda_1(\mathcal{L}) \quad \text{and} \quad \|b_1\|_2 \leq 2^{(d-1)/4} \det(\mathcal{L})^{1/d},$$

*so it solves  $\gamma$ -SVP for  $\gamma = 2^{(d-1)/2}$  and guarantees  $\delta_0 = 2^{1/4} \approx 1.1892$ , achieving about 1.0219 in practice.*

### 6.3 Example

**Example 10** (Reduction in miniature: the bad basis repaired). Run Lagrange–Gauss on the bad basis  $b_1 = (2, 3)$ ,  $b_2 = (4, 9)$  of Example 6, which has defect  $\approx 5.918$ .

$$\begin{aligned} b_1 = (2, 3), \quad b_2 = (4, 9) : \quad \mu &= \frac{8+27}{13} = \frac{35}{13} \approx 2.692, \quad \lfloor \mu \rfloor = 3, \quad b_2 \leftarrow (4, 9) - 3(2, 3) = (-2, 0); \\ b_1 = (-2, 0), \quad b_2 = (2, 3) : \quad \mu &= \frac{-4}{4} = -1, \quad \lfloor \mu \rfloor = -1, \quad b_2 \leftarrow (2, 3) - (-2, 0) = (0, 3). \end{aligned}$$

After the first step  $\|(-2, 0)\|_2 = 2 < \sqrt{13} = \|(2, 3)\|_2$ , so the two vectors are swapped; after the second,  $\|(0, 3)\|_2 = 3 \geq 2$  and the algorithm stops. The output is  $\{(-2, 0), (0, 3)\}$ : the good basis of Example 6 up to sign, with inner product 0, determinant 6 ✓, defect  $2 \cdot 3/6 = 1$ , and first vector of length  $\lambda_1 = 2$ . Two steps turned 5.918 into 1.

**Example 11** (Two evaluations of  $\delta_0(\beta)$ ). With the estimate above,

$$\delta_0(100) \approx 1.00926 \quad \text{at classical cost } 2^{0.292 \cdot 100} = 2^{29.2}, \quad \delta_0(400) \approx 1.00398 \quad \text{at } 2^{116.8},$$

quantumly  $2^{26.5}$  and  $2^{106.0}$ . Moving  $\delta_0$  by one unit in the third decimal place costs a factor of about  $2^{116.8-29.2} = 2^{87.6}$  in work.

## 6.4 What Parameter Selection Actually Is

A designer picks  $(n, q, \sigma)$  so that the cheapest block size at which BKZ would find the scheme's planted short vector already costs more than the target security level. Every parameter table in this course is the output of that loop.

**Definition 10** (The parameter-selection loop).

1. Fix the scheme's shape. This determines a  $q$ -ary lattice (Section 7) of dimension  $d$  and determinant  $q^k$  or  $q^{d-k}$ , carrying a planted vector — the secret, or the error — of expected norm about  $\sigma\sqrt{d}$ .
2. Compute  $\det(\mathcal{L})^{1/d}$  and the Gaussian-heuristic length  $\sqrt{d/2\pi e} \det(\mathcal{L})^{1/d}$  of a generic lattice vector, and check that the planted vector is shorter.
3. Find the root Hermite factor  $\delta_0$  an attacker needs for BKZ's output to reach the planted vector before it reaches that generic length.
4. Invert  $\delta_0(\beta)$  for the block size  $\beta$ , and cost it:  $2^{0.292\beta}$  classically,  $2^{0.265\beta}$  quantumly.
5. If that cost is below  $2^\lambda$ , raise  $d$  (that is,  $n$  or the module rank  $k$ ), lower  $q$ , or raise  $\sigma$ , and return to step 1.

**Example 12** (One parameter set, costed). *Steps 4 and 3, backwards.* For 128 bits of classical security under the core-SVP model we need  $0.292\beta \geq 128$ , i.e.  $\beta \geq 438.36$ , so  $\beta = 439$ . Feeding that into the estimate of Section 6.2,

$$\delta_0(439) = \left( \frac{439}{2\pi e} (439\pi)^{1/439} \right)^{1/876} \approx 1.003732,$$

against LLL's 1.1892.

*Step 2, forwards.* At ML-KEM-512 scale,  $d = 512$  and determinant  $q^{256}$  with  $q = 3329$  give

$$\det(\mathcal{L})^{1/d} = 3329^{256/512} = \sqrt{3329} \approx 57.70, \quad \text{gh}(\mathcal{L}) \approx 5.475 \cdot 57.70 \approx 315.9,$$

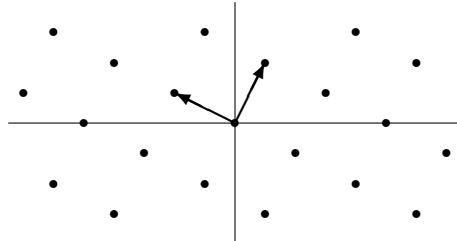
against an error vector of expected norm  $3.2\sqrt{512} \approx 72.41$ : the planted vector is  $315.9/72.41 \approx 4.36$  times shorter than generic.

*Step 5.* ML-KEM-512 is published at  $\beta \approx 406$ , i.e.  $0.292 \cdot 406 \approx 118.6$  bits of core-SVP cost.

## 7. $q$ -ary Lattices: The Missing Link

### 7.1 Two Lattices from One Matrix

A matrix  $A$  over  $\mathbb{Z}_q$  gives two lattices of integer vectors: the ones it kills,  $Az \equiv 0 \pmod{q}$ , and the ones it produces, the integer vectors congruent modulo  $q$  to a combination of its rows. Both are unions of residue classes modulo  $q$ , so each looks like a sheared copy of  $\mathbb{Z}^d$  thinned out by a factor that is a power of  $q$ .



The dots are  $\Lambda_5^\perp((1 \ 2))$ , the integer pairs  $(z_1, z_2)$  with  $z_1 + 2z_2 \equiv 0 \pmod{5}$ : one point in every five of  $\mathbb{Z}^2$ , so the determinant is  $5 = q^k$ . The two arrows are  $(1, 2)$  and  $(-2, 1)$ , both shortest vectors, of length  $\sqrt{5}$ .

**Definition 11** ( $q$ -ary lattices). Fix a prime  $q$ , integers  $k < d$ , and  $A \in \mathbb{Z}_q^{k \times d}$  of full rank  $k$  modulo  $q$ . Set

$$\Lambda_q(A) = \{y \in \mathbb{Z}^d : y \equiv A^T s \pmod{q} \text{ for some } s \in \mathbb{Z}^k\}, \quad \Lambda_q^\perp(A) = \{z \in \mathbb{Z}^d : Az \equiv 0 \pmod{q}\}.$$

A lattice is  **$q$ -ary** when  $q\mathbb{Z}^d \subseteq \mathcal{L} \subseteq \mathbb{Z}^d$ , i.e. when membership depends only on residues modulo  $q$ ; both of these are, and both are therefore full-rank in  $\mathbb{R}^d$ .

**Proposition 2.**  $\det \Lambda_q(A) = q^{d-k}$  and  $\det \Lambda_q^\perp(A) = q^k$ . Moreover, writing  $A = [I_k \mid \bar{A}]$  after a column permutation, the columns of  $\begin{pmatrix} I_k & 0 \\ \bar{A}^T & qI_{d-k} \end{pmatrix}$  generate  $\Lambda_q(A)$  and those of  $\begin{pmatrix} -\bar{A} & qI_k \\ I_{d-k} & 0 \end{pmatrix}$  generate  $\Lambda_q^\perp(A)$ , and  $\Lambda_q^\perp(A) = q \cdot \Lambda_q(A)^*$ .

By Proposition 1: the map  $z \mapsto Az \pmod{q}$  from  $\mathbb{Z}^d$  onto  $\mathbb{Z}_q^k$  is surjective with kernel  $\Lambda_q^\perp(A)$ , of index  $q^k$ ; and  $\Lambda_q(A)/q\mathbb{Z}^d$  is the row space of  $A$ , of size  $q^k$ , so  $\Lambda_q(A)$  has index  $q^d/q^k = q^{d-k}$ .

*Example.*

**Example 13** (Both lattices of one small matrix). Keep  $q = 13$ ,  $k = 1$ ,  $d = 3$  and  $A = (1 \ 5 \ 8)$  from Example 9.

*The kernel lattice.*  $\Lambda_{13}^\perp(A)$  is generated by the columns

$$\begin{pmatrix} -5 & -8 & 13 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad \det = 13 = q^k \checkmark$$

(expand along the second row). Each generator satisfies the congruence:  $-5 + 5 = 0$ ,  $-8 + 8 = 0$ ,  $13 \equiv 0$ . Its shortest vector is  $\lambda_1 = \|(0, 1, 1)\|_2 = \sqrt{2}$ . The Gaussian heuristic here is  $13^{1/3}/(4\pi/3)^{1/3} \approx 2.351/1.612 \approx 1.459$ , within about 3% of the truth even at  $d = 3$ ; Minkowski gives  $\sqrt{3} \cdot 13^{1/3} \approx 4.073$ .

*The image lattice.*  $\Lambda_{13}(A)$  is generated by  $(1, 5, 8)$ ,  $(0, 13, 0)$ ,  $(0, 0, 13)$ , i.e.

$$\begin{pmatrix} 1 & 0 & 0 \\ 5 & 13 & 0 \\ 8 & 0 & 13 \end{pmatrix}, \quad \det = 1 \cdot 13 \cdot 13 = 169 = q^{d-k} \checkmark$$

Exhaustive search gives  $\lambda_1(\Lambda_{13}(A)) = \sqrt{17} \approx 4.123$ , attained at  $\pm(3, 2, -2)$ , which is  $3 \cdot (1, 5, 8) = (3, 15, 24)$  reduced to centred representatives modulo 13.

## 7.2 An LWE Sample Is a BDD Instance

An LWE sample is a point of the image lattice nudged by a short error vector, so recovering the secret means finding the lattice point nearest a target that is known to lie close to the lattice.

**Proposition 3.** Let  $A \in \mathbb{Z}_q^{k \times d}$ ,  $s \in \mathbb{Z}_q^k$ ,  $e \in \mathbb{Z}^d$  short, and  $b = A^T s + e \pmod{q}$ . Then  $b$  lies within  $\|e\|_2$  of  $\Lambda_q(A)$ , and if

$$\|e\|_2 < \frac{1}{2} \lambda_1(\Lambda_q(A))$$

the closest lattice point to  $b$  is unique and determines  $s$  and  $e$ . So search-LWE is BDD on  $\Lambda_q(A)$ , and decision-LWE asks whether  $b$  is close to  $\Lambda_q(A)$  or uniform in  $\mathbb{Z}_q^d$ . With  $e = 0$  the problem is Gaussian elimination over  $\mathbb{Z}_q$ .

*Example.*

**Example 14** (An LWE sample as a BDD instance, checkable by hand). Same  $q = 13$ ,  $A = \begin{pmatrix} 1 & 5 & 8 \end{pmatrix}$ , and the lattice  $\Lambda_{13}(A)$  of determinant 169 and  $\lambda_1 = \sqrt{17}$  from Example 13. Take the secret  $s = 2$ :

$$A^T s = (2, 10, 16) \equiv (2, -3, 3) \pmod{13},$$

a genuine lattice point in centred representatives. Add the short error  $e = (1, -1, 1)$ :

$$b = (3, -4, 4) \equiv (3, 9, 4) \pmod{13}.$$

The BDD promise holds, since  $\|e\|_2 = \sqrt{3} \approx 1.732$  and  $\frac{1}{2}\lambda_1 = \frac{1}{2}\sqrt{17} \approx 2.062$ . The closest lattice point is  $(2, -3, 3)$  at distance  $\sqrt{3}$ ; the runner-up is  $(4, -6, 6)$ , which is  $s = 4$ , at distance 3. Finding the closest vector recovers  $s = 2$  and  $e = (1, -1, 1)$  together. Had the error been three times larger,  $\|e\|_2 = 3\sqrt{3} \approx 5.196$  would exceed  $\lambda_1$  itself and the answer would no longer be unique.

### 7.3 SIS Is SVP

An SIS solution is a short nonzero vector of the kernel lattice, so SIS is the shortest-vector problem with a norm bound in place of exactness.

**Proposition 4.**  $\{z \in \mathbb{Z}^d : Az \equiv 0 \pmod{q}\}$  is  $\Lambda_q^\perp(A)$  by definition, so SIS with parameters  $(q, k, d, \beta)$  is the problem of finding a nonzero vector of norm at most  $\beta$  in  $\Lambda_q^\perp(A)$ : that is  $\gamma$ -SVP on  $\Lambda_q^\perp(A)$  with  $\gamma = \beta/\lambda_1(\Lambda_q^\perp(A))$ , whose decision version is GapSVP.

*Example.*

**Example 15** (One computation stated twice). With  $q = 13$  and  $A = \begin{pmatrix} 1 & 5 & 8 \end{pmatrix}$ , the vector  $z = (0, 1, 1)$  of Example 9 is the SIS solution of norm  $\sqrt{2}$ , and the kernel half of Example 13 found the same  $(0, 1, 1)$  as the shortest vector of  $\Lambda_{13}^\perp(A)$ , a lattice of determinant  $q^k = 13$ . Hence  $\lambda_1 = \sqrt{2}$  and the instance is  $\gamma$ -SVP with  $\gamma = \beta/\sqrt{2}$ : at  $\beta = \sqrt{2}$  it is exact SVP, at  $\beta = 13$  it is  $\gamma \approx 9.19$  and trivial.

## 8. Ideal and Module Lattices

An ideal lattice is a lattice with extra symmetry inherited from the ring: multiplying a ring element by  $X$  moves its coefficient vector to another vector of the same lattice and of exactly the same length. Stacking  $k$  copies of the ring gives a module lattice, which has the same symmetry in each block.

**Definition 12** (Coefficient embedding, ideal and module lattices). The **coefficient embedding** of  $R = \mathbb{Z}[X]/(X^n + 1)$  is

$$\iota : R \rightarrow \mathbb{Z}^n, \quad a_0 + a_1X + \cdots + a_{n-1}X^{n-1} \mapsto (a_0, a_1, \dots, a_{n-1}),$$

an isomorphism of additive groups carrying  $\|\cdot\|_\infty$  on  $R$  to  $\|\cdot\|_\infty$  on  $\mathbb{Z}^n$ . The image  $\iota(I)$  of an ideal  $I \subseteq R$  is an **ideal lattice**. Applied coordinatewise,  $\iota$  sends  $R^k$  to  $\mathbb{Z}^{nk}$ , so a rank- $k$  module over a degree- $n$  ring is a lattice of dimension  $d = nk$  (for ML-KEM-768,  $n = 256$  and  $k = 3$  give  $d = 768$ ).

**Proposition 5.** In  $\mathbb{Z}[X]/(X^n+1)$ , multiplication by  $X$  acts on coefficient vectors as the *negacyclic shift*

$$X \cdot (a_0, a_1, \dots, a_{n-1}) = (-a_{n-1}, a_0, a_1, \dots, a_{n-2}),$$

the sign appearing because  $X \cdot X^{n-1} = X^n = -1$ . It permutes coordinates and flips one sign, so it is an isometry for both  $\|\cdot\|_2$  and  $\|\cdot\|_\infty$ , and an ideal lattice is closed under it. Hence one short vector of an ideal lattice yields  $n$  of them, pairwise independent and all of the same length.

### 8.3 Example

**Example 16** (The orbit of one element, and its determinant). In  $R = \mathbb{Z}[X]/(X^4 + 1)$  take  $a = 1 - X + 2X^3$ , i.e.  $(1, -1, 0, 2)$ . Shifting negacyclically three times, each shift moving every entry one place to the right and bringing the last one back negated:

$$\iota(a) = (1, -1, 0, 2), \quad \iota(Xa) = (-2, 1, -1, 0), \quad \iota(X^2a) = (0, -2, 1, -1), \quad \iota(X^3a) = (1, 0, -2, 1),$$

all four of Euclidean norm  $\sqrt{1+1+0+4} = \sqrt{6} \approx 2.449$  and infinity norm 2  $\checkmark$ . As columns they form a basis matrix of the ideal lattice  $\iota(aR)$ , whose determinant is 18; so  $\iota(aR)$  has index 18 in  $\mathbb{Z}^4$ . That 18 is the field norm of  $a$ : evaluating at the four primitive 8-th roots of unity  $\zeta^j$ ,  $j \in \{1, 3, 5, 7\}$  — the evaluation points of Lecture 3’s negacyclic NTT — gives  $\prod_j a(\zeta^j) = 18$ .  $\checkmark$

## Summary

1. A lattice is a discrete additive subgroup of  $\mathbb{R}^d$ : the integer span of  $k$  independent vectors, rank  $k$ , dimension  $d$ . Two bases generate it exactly when they differ by a unimodular  $U$ , so every lattice of rank  $\geq 2$  has infinitely many bases, and  $\det(\mathcal{L}) = |\det B| = \text{vol}(\mathcal{P}(B))$  is basis-independent. For  $\mathcal{L} \subseteq \mathbb{Z}^d$  it is the index  $[\mathbb{Z}^d : \mathcal{L}]$ , so  $k$  congruences modulo  $q$  give  $\det = q^k$ .
2. Gram–Schmidt makes quality numerical: the product of the GSO norms is pinned to  $\det(\mathcal{L})$ , so the defect  $\prod_i \|b_i\|_2 / \det(\mathcal{L}) \geq 1$  measures the gap. The two bases of  $2\mathbb{Z} \times 3\mathbb{Z}$  used here have defects 1 and 5.918, decoding radii 1 and 0.832; a good basis kept secret behind a published bad one is a lattice trapdoor.
3. Minkowski’s bound  $\lambda_1 \leq \sqrt{d} \det(\mathcal{L})^{1/d}$  holds unconditionally, while the Gaussian heuristic  $\lambda_1 \approx \sqrt{d/2\pi e} \det(\mathcal{L})^{1/d}$  is sharper in large dimension and unreliable in small.
4. SVP, CVP and BDD, with factor  $\gamma$  and decision form GapSVP, make a ladder: exact is NP-hard,  $\gamma = 2^{O(d)}$  is easy by LLL, and cryptography lives at  $\gamma = \text{poly}(d)$  between. SIS asks for short nonzero  $z$  with  $Az \equiv 0$ ; both adjectives are load-bearing, and its hardness is collision resistance of  $z \mapsto Az$ .
5. LLL is polynomial with  $\|b_1\|_2 \leq 2^{(d-1)/2} \lambda_1$ ; BKZ with block size  $\beta$  interpolates to exact SVP at cost  $2^{0.292\beta}$ . Quality is the root Hermite factor  $\delta_0$  in  $\|b_1\|_2 = \delta_0^d \det(\mathcal{L})^{1/d}$ , and choosing  $(n, q, \sigma)$  so the necessary  $\beta$  costs over  $2^\lambda$  is parameter selection.
6.  $q$ -ary lattices join the halves: LWE  $(A, b = A^T s + e)$  is BDD on  $\Lambda_q(A)$ , determinant  $q^{d-k}$ ; SIS is SVP on  $\Lambda_q^\perp(A)$ , determinant  $q^k$ .
7. The coefficient embedding turns a rank- $k$  module over a degree- $n$  ring into a lattice of dimension  $nk$ , and multiplication by  $X$  is a norm-preserving negacyclic shift, so short vectors come in orbits of size  $n$ . Structure buys small keys and  $O(n \log n)$  arithmetic at the price of a smaller, less-understood instance class.

## Exercises

1. Verify that  $U = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}$  is unimodular and compute  $U^{-1}$ ; then show that  $\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$  does *not* generate  $2\mathbb{Z} \times 3\mathbb{Z}$ , and say which points are missing.
2. Let  $\mathcal{L}$  have basis  $(3, 0), (1, 2)$ . Compute  $\det(\mathcal{L})$  and  $[\mathbb{Z}^2 : \mathcal{L}]$ , then give congruence conditions on  $(u, v)$  equivalent to membership and check them against the index.
3. Do Gram–Schmidt in exact fractions on  $(1, 1, 0), (1, 0, 1), (0, 1, 1)$ . Verify  $\prod_i \|\tilde{b}_i\|_2 = |\det B| = 2$ , compute the defect and  $\lambda_1$ , and check both Minkowski’s bound and the Gaussian heuristic.
4. For  $q = 13$  and  $A = \begin{pmatrix} 1 & 4 \end{pmatrix}$ , write a basis of  $\Lambda_q^\perp(A)$ , check its determinant against Proposition 2, and run Lagrange–Gauss by hand, reporting  $\lambda_1$  and the defect before and after.
5. With  $q = 13$  and  $A = \begin{pmatrix} 1 & 5 & 8 \end{pmatrix}$ , find every solution of  $Az \equiv 0$  with  $\|z\|_2 \leq 2$ . What goes wrong with SIS if  $\beta \geq q$  is allowed, and what if  $z = 0$  is?
6. With the same  $q$  and  $A$ , let  $b = (3, 9, 4)$  be an LWE sample. Verify  $\det \Lambda_q(A)$ , find the closest lattice vector in centred representatives, read off  $s$  and  $e$ , and confirm the BDD promise.
7. In  $R = \mathbb{Z}[X]/(X^4 + 1)$  let  $a = 2 + X - X^2$ . Write the coefficient vectors of  $a, Xa, X^2a, X^3a$ , check the four norms agree, and compute their determinant. Why does this hand an attacker  $n$  short vectors for the price of one?
8. With  $\delta_0(\beta) = \left(\frac{\beta}{2\pi e}(\pi\beta)^{1/\beta}\right)^{1/(2(\beta-1))}$ , compute  $\delta_0$  for  $\beta = 100, 200, 400$  and find the least  $\beta$  with  $0.292\beta \geq 128$ . Why does a factor-of-two cost-model error move  $\beta$  by only a few units?